# Zeqi Li

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## Research Interests

Embodied AI; foundation models (LLMs/VLMs/VLAs) for robot reasoning and planning; active object search and navigation with mobile manipulators; forceful bimanual and multi-arm manipulation; humanoid and multi-robot systems.

## Education

**Technical University of Munich** Munich, Germany
*M.Sc. in Robotics, Cognition, Intelligence* Oct 2022 – Feb 2025
**GPA:** 3.6 / 4.0 (very good) **Thesis Grade:** 4.0 / 4.0 (very good)
**Thesis:** *Integrating Large Language Models with Mobile Manipulator for Enhanced Target Search*

- Proposed a collaborative reasoning framework integrating large language models (LLMs) with mobile manipulators to improve search efficiency and localization accuracy in unknown scenes.
- Designed a hierarchical search strategy combining semantic-spatial dynamic coupling, leveraging advanced simulators and physics engines for validation.

**University of Siegen** Siegen, Germany
*B.Sc. in Electrical Engineering – Bachelor Phase II* Oct 2020 – Mar 2022
**GPA:** 3.5 / 4.0 (very good) **Thesis Grade:** 3.7 / 4.0 (very good)
**Thesis:** *Implementation and comparison of two variants of the Kalman Filter for estimating the side-slip angle of a vehicle within the simulation environment MATLAB/Simulink*

- Developed and simulated Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF) algorithms for vehicle side-slip angle estimation in MATLAB/Simulink.
- Conducted performance optimization analysis and proposed reproducible solutions to enhance reliability of vehicle state estimation systems.

**Chinese-German Technical Faculty, Qingdao University of Science and Technology** Qingdao, China
*B.Sc. in Automation – Bachelor Phase I* Sep 2017 – Sep 2020
**Transferred GPA:** 3.6 / 4.0

## Publications

***Peer-reviewed***

[1] L. Zhang*, **Z. Li***, K. Cai*, Q. Huang, Z. Bing, A. Knoll. "Language-Enhanced Mobile Manipulation for Efficient Object Search in Indoor Environments." *IEEE International Conference on Cyborg and Bionic Systems (CBS)*, 2025. doi:10.1109/CBS65871.2025.11267741 • arXiv:2508.20899

[2] K. Cai*, C. Wang*, **Z. Li**, H. Yao, W. Chen, L. Figueredo, A. Billard, A. Ajoudani. "Imitation-Guided Bimanual Planning for Stable Manipulation under Changing External Forces." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. doi:10.1109/IROS60139.2025.11246712 • arXiv:2509.19261

[3] Q. Tong, K. Cai, **Z. Li**, O. Nocentini, M. Lagomarsino, M. Lorenzini, C. Wang, A. Ajoudani. "From geometry to intention: A survey of Head Pose Estimation for HCI and HRI systems." *Biomimetic Intelligence and Robotics*, 2026, 100360. doi:10.1016/j.birob.2026.100360

***Under review***

Two conference papers at IEEE ICRA 2027; one first-author journal paper at *The International Journal of Robotics Research (IJRR)*.

***Ongoing***

Two first-author papers in preparation, targeting *CVPR 2027* and *RSS 2027*.

## Research Experience

**Italian Institute of Technology (IIT)** Genoa, Italy
*Remote Research Assistant* May 2025 – Current
**Papers:** Survey on head pose estimation for HCI/HRI (Biomimetic Intelligence and Robotics, 2026) — Publication [3]; first-author journal paper under review at IJRR
*Research Collaborator* Jan 2025 – Apr 2025
**Paper:** *Imitation-Guided Bimanual Planning for Stable Manipulation under Changing External Forces* (third author, IROS 2025) — Publication [2]

- Proposed an imitation-guided bimanual planning framework to enhance robotic grasping stability and dexterity in dynamic environments.
- Implemented a hierarchical two-stage motion architecture with stable grasp manifold intersection sampling, improving motion performance under variable external forces.
- Demo Video: https://youtu.be/ZUdxyTh7uUI

**KI.Fabrik** Munich, Germany
*Research Intern* May 2024 – Feb 2025
**Paper:** *Language-Enhanced Mobile Manipulation for Efficient Object Search in Indoor Environments* (co-first author, IEEE CBS 2025) — Publication [1]

- Designed a language model-driven hierarchical navigation framework for robotic object search in unknown scenes.
- Integrated structured prompt-guided commonsense reasoning with hybrid motion planning (polar angle sorting + distance prioritization), improving search success rate and path efficiency.
- Demo Video: https://youtu.be/e0FzMS5M7dw

## Teaching & Academic Service

**University of Siegen** Siegen, Germany
*Academic Assistant* Apr 2022 – Sep 2022

- Conducted tutorial sessions for courses including *Circuits, Electromagnetics, and Signal Systems*, delivering 108 instructional hours with consistent positive feedback.
- Developed course materials, problem sets, and guided students through complex technical concepts.

**University of Siegen** Siegen, Germany
*Student Assistant* Sep 2021 – Dec 2021

- Mentored international freshmen on academic integration and daily life in Germany.
- Collaborated with the professor to design and refine course content, ensuring alignment with curriculum goals.

## Honors & Awards

**National Scholarship** Dec 2019
**Outstanding Student of Shandong Province** Aug 2020

## Skills

**Language:** Chinese (native); English (M.Sc. taught in English); German (B.Sc. taught in German)
**Programming:** Python/PyTorch, C/C++, MATLAB/Simulink, Git
**Robotics:** Linux OS, ROS, CoppeliaSim, NVIDIA Isaac Sim, SolidWorks
**Design:** LaTeX, HTML, Adobe PS/PR/DC**,** MS Office, PLC, Blender